# Bell Programmer Manual

modules, examples, structure file

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# 1 Programmer Manual

Physics is in the Beam Physics Manual. Architecture of the blocks is in Architecture. This page lists entry points of the Julia package `Bell`.

The object of the ring is the Lie product of Dragt–Finn factors. Concatenate in the algebra, read the product, track a point through it. Linear optics is the closed form of $\exp(: f_2 :)$, stored as the matrix $R$. It is not a second theory of the lattice.

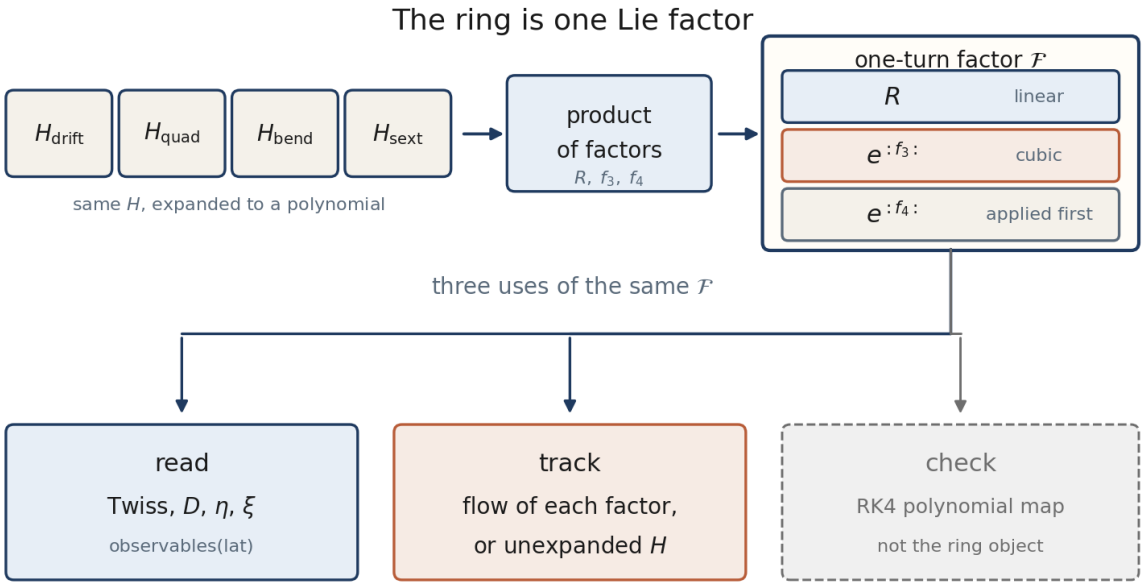



Figure 1: The ring is one Lie factor: Hamiltonians are concatenated in the algebra; the product is read, tracked, or checked.

`observables(lat)` builds `ring_lie_factor` once and reads Twiss, $\eta$ and $\xi$ from that factor. `ring_taylor_map` is an RK4 polynomial and is not that object.

## 1.1 Layout

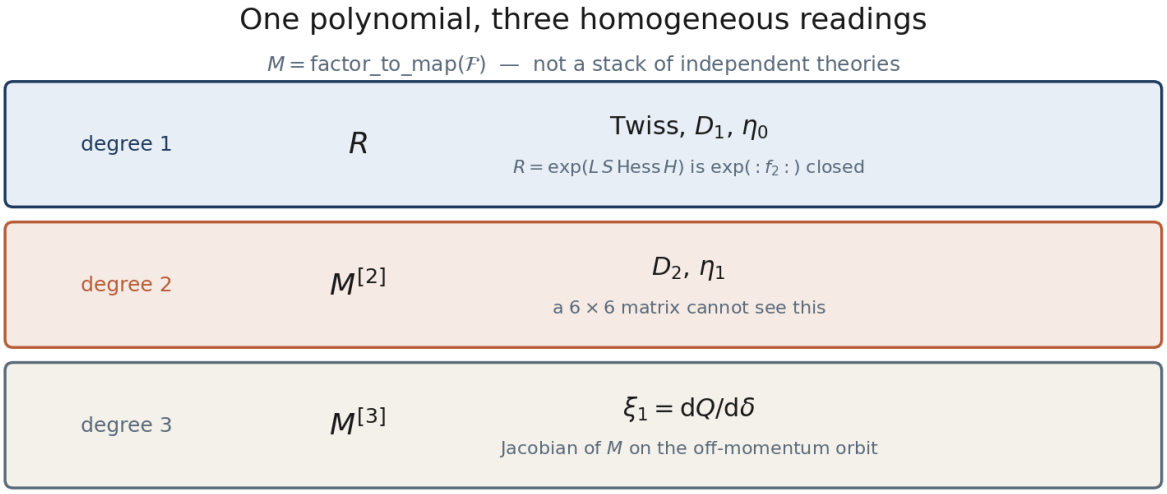



Figure 2: One polynomial, three homogeneous readings: degree 1 gives Twiss and $\eta_0$; degree 2 gives $D_2$ and $\eta_1$; degree 3 gives $\xi_1$.

| **path** | **role** |
|---|---|
| `src/coordinates.jl` | particle, reference, $v(\delta)$, $e(\delta)$, $z \leftrightarrow u$ |
| `src/elements.jl` | element types (data only) |
| `src/structure.jl` | structure file, `Lattice`, sequence edits |
| `src/hamiltonians.jl` | element generator $H$ as a polynomial |
| `src/lie_algebra.jl` | polynomial space, Poisson bracket, Dragt–Finn product |
| `src/optim_converter.jl` | writes a structure file from OptiM *.opt* |
| `src/linear.jl` | $R = \exp(LS\mathrm{Hess}\, H)$, slice Twiss along $s$ |
| `src/symplectic.jl` | $J^{\mathsf{T}}SJ - S$ |
| `src/tracking.jl` | flow of each factor, thin RF kick, product of the ring |
| `src/higher_order.jl` | closed orbit, $D_2$, $\eta$, chromaticity, `observables` |
| `src/differential_algebra.jl` | independent DA map (RK4 in polynomials) — a check |
| `src/normal_form.jl` | one-turn factor with thin RF; torus average of $\delta$ |
| `src/longitudinal.jl` | averaged longitudinal Hamiltonian in $(\varphi, \delta)$ |
| `src/exact_integrator.jl` | unexpanded $H$, finite cavity, renormalise |
| `src/acceleration.jl` | ramp, tune fit |
| `src/plotting.jl` | figures |
| `src/optimizers.jl` | optional sextupole response |
| `src/Bell.jl` | loads every block and re-exports |

Entry: from the `Julia` folder, `using Bell`.

The one-turn object used for the normal form is `one_turn_factor_rf`. Optics of the magnetic product without RF is `observables(lat)`.

## 1.2 Examples

```
julia --project=. examples/10_linear_optics.jl
julia --project=. examples/11_lie_maps.jl
julia --project=. examples/12_higher_order.jl
julia --project=. examples/13_optional_sextupole_optimization.jl
julia --project=. examples/14_rf_acceleration.jl
julia --project=. examples/15_coordinates.jl
julia --project=. examples/16_blond_longitudinal.jl
```

```
julia --project=. examples/17_lie_product.jl
julia --project=. examples/18_lie_normal_form.jl
```

The equilibrium-momentum study lives in `C:\GIT_REPOS\SCT_2\Julia\equilibrium_energy`. It depends on this package via `Pkg.develop` of the `Julia` folder.

### 1.3 Accepted design choices

- Injection kinetic energy 10 MeV; final 270 MeV from the lattice header. - Stationary study: thin RF, $V = 0.2$ MV, $h = 5$, $\varphi_s = \pi$. - Ramp demo: finite cavity $L = 1$ m, $\varphi_s = 150^\circ$. - Mass = CODATA deuteron 1875.61294257 MeV (OptiM `Mass=2*938.272` ignored). - Lattice sequence used once (file `Number of periods=2` is an error; bend is 360 deg). - Closed $D(0) = 0.011$ m (OptiM initial 0 is not the closed solution). - Default sextupole strengths are zero.

### 1.4 Docs build

```
julia --project=docs docs/make.jl
```

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